MINI PROJECT

GLOBAL ELECTRIC VEHICLE ANALYTICS

EXP.NO-9 | Data Analytics / Sustainable Mobility

AIM

To clean, analyze, and visualize electric vehicle data using Python, Power BI, and Tableau to identify adoption trends, regional differences, sales growth, and charging-infrastructure patterns.

DOMAIN

Electric Vehicle Analytics / Sustainable Mobility Analytics

PROBLEM STATEMENT

Electric vehicle adoption is changing rapidly across countries and regions. Sales, vehicle stock, market share, and charging infrastructure provide different perspectives on this transition. This project organizes published EV data into an analytical dataset and uses visualizations to compare adoption patterns over time and across regions.

DATASET DESCRIPTION

The proposed primary source is the International Energy Agency (IEA) Global EV Data Explorer and its downloadable Global EV Outlook dataset. The source provides historical and projected indicators related to electric-vehicle sales, stock, charging infrastructure, and related measures. The project will use historical observations for analysis and will keep projections clearly distinguished from observed data.

Planned fields include: year, country or region, indicator, vehicle category where available, value, and unit. The exact row count and column count will be recorded after downloading and inspecting the source file.

Source: IEA Global EV Data Explorer, updated 20 May 2026. Data licence and attribution requirements should be checked on the downloaded file's source page.

PROCEDURE

• Download the relevant historical EV dataset from the IEA Global EV Data Explorer.

• Load the downloaded CSV/XLSX data using Python and Pandas.

• Inspect the dataset structure, column names, data types, missing values, and duplicate records.

• Remove exact duplicate rows where appropriate and standardize country, region, indicator, and unit labels.

• Check missing values and distinguish genuine zero values from unavailable observations.

• Filter to historical observations for trend analysis and separate projections if included.

• Create derived fields such as year-on-year change and year-on-year growth percentage where consecutive observations are available.

• Perform exploratory data analysis using Pandas.

• Create static visualizations with Matplotlib and Seaborn and an interactive chart with Plotly.

• Export the cleaned dataset for use in Power BI and Tableau.

• Build an interactive Global EV Intelligence dashboard in Power BI.

• Build a complementary EV Adoption Explorer dashboard in Tableau.

KEY ANALYSIS QUESTIONS

• How have global electric-car sales changed over the selected historical period?

• Which countries or regions record the highest sales and vehicle stock?

• How does EV market share vary across countries and years?

• How has public charging infrastructure changed over time?

• Which regions show different relationships between EV stock and public charging points?

Important reporting note: final numerical findings must be generated from the exact downloaded dataset. The charts in this report are source-aligned overview illustrations, not a substitute for running the full cleaning and analysis workflow.

PYTHON ANALYSIS

Python will be used for data inspection, cleaning, processing, exploratory data analysis, and visualization. Pandas will handle tabular operations, NumPy will support calculations, and Matplotlib, Seaborn, and Plotly will be used for charts.

DATA CLEANING

The raw file will be inspected for missing values, duplicate rows, inconsistent labels, and unexpected data types. Indicator names and units must be retained accurately because sales, stock, and market-share measures represent different concepts. Missing values will not automatically be replaced with medians, since missing country-year observations may not be comparable or statistically interchangeable.

DATA PROCESSING

Potential derived fields include:

Year-on-Year Change = Current-year value – Previous-year value

Year-on-Year Growth (%) = ((Current-year value – Previous-year value) / Previous-year value) × 100

EV Market Share = Electric-car sales / Total car sales × 100, only if a compatible total-car-sales field is available.

Growth calculations will be performed within each country, region, and indicator series. Rows with missing prior-year values or a zero denominator will be handled explicitly.

EXPLORATORY DATA ANALYSIS

• Annual global EV sales trend

• Country and regional sales comparison

• Electric-car stock by country or region

• EV sales-share comparison over time

• Public charging infrastructure trend

• Relationship between EV stock and public charging points

PYTHON VISUALIZATIONS

Planned visualizations include a line chart for annual sales, horizontal bars for country rankings, a heatmap for country-year comparisons, and an interactive Plotly chart for exploring regional indicators. Final plots will be generated from the cleaned source file.

PYTHON VISUALIZATIONS

The following overview charts summarize selected public IEA findings and demonstrate the intended visualization style. Values are rounded headline figures or approximate values read from published charts. They should not be treated as the exact output of the final CSV/XLSX analysis.

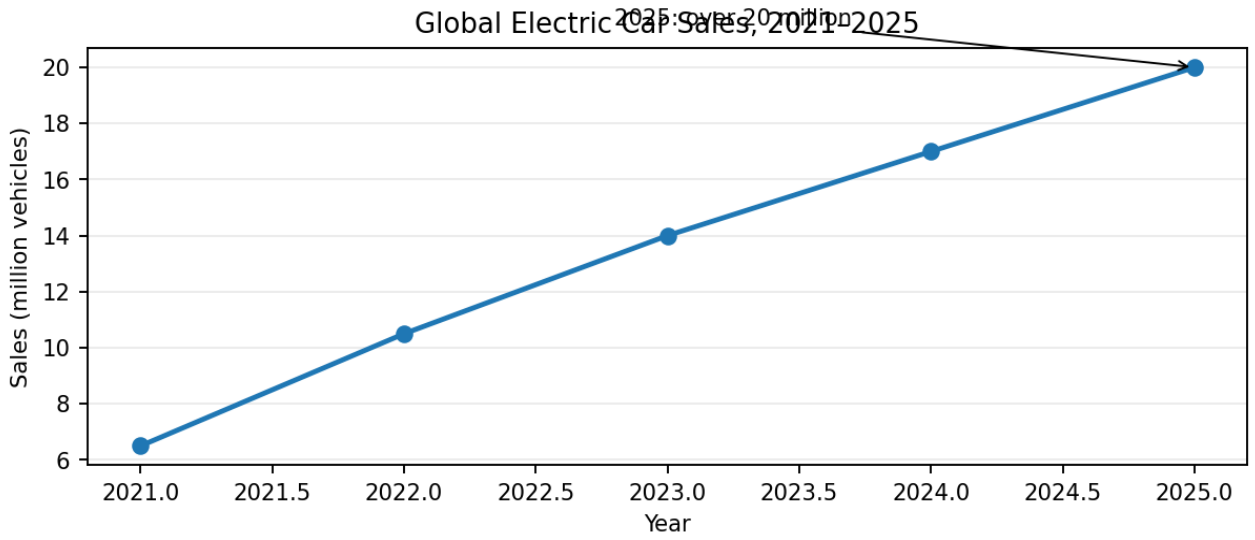


Figure 1. Global electric-car sales increased substantially between 2021 and 2025. The 2025 headline represents more than 20 million electric cars sold worldwide.

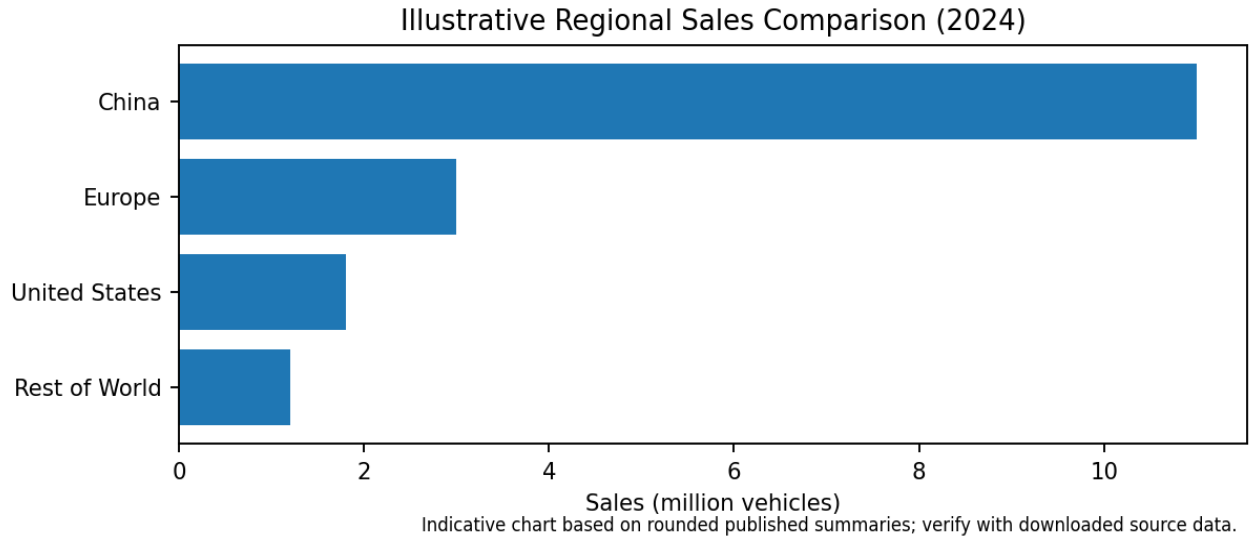


Figure 2. Regional comparison concept. Exact regional values and category definitions must be confirmed against the downloaded dataset before submission.

ADDITIONAL EV INDICATORS

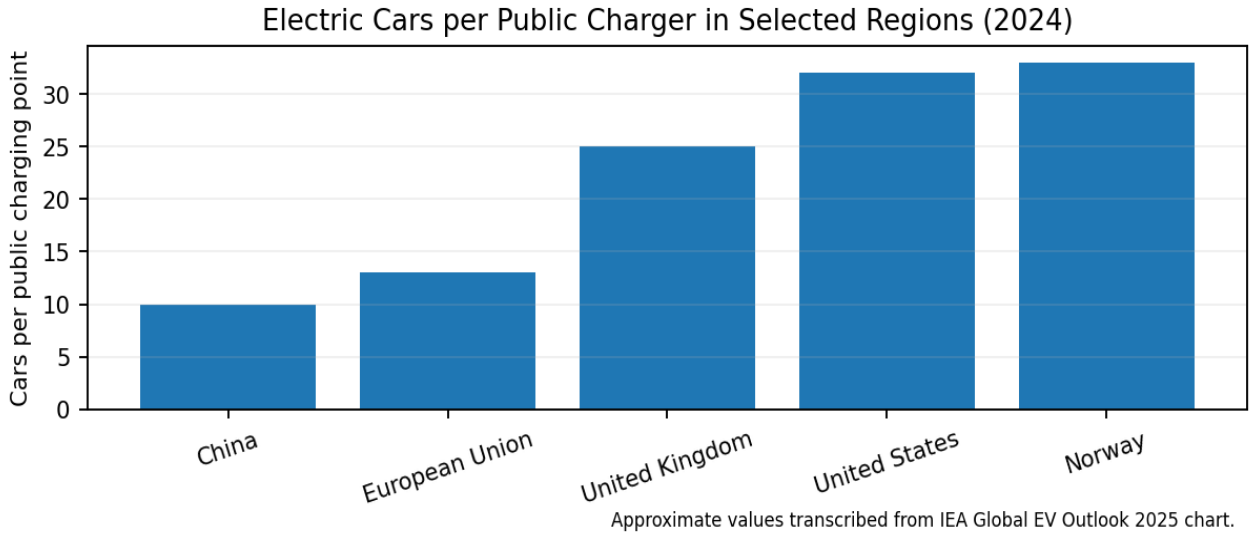


Figure 3. Approximate number of electric cars per public charging point in selected regions, based on the IEA Global EV Outlook 2025 published chart. Charging demand and access differ across regions, so this ratio should be interpreted alongside local context.

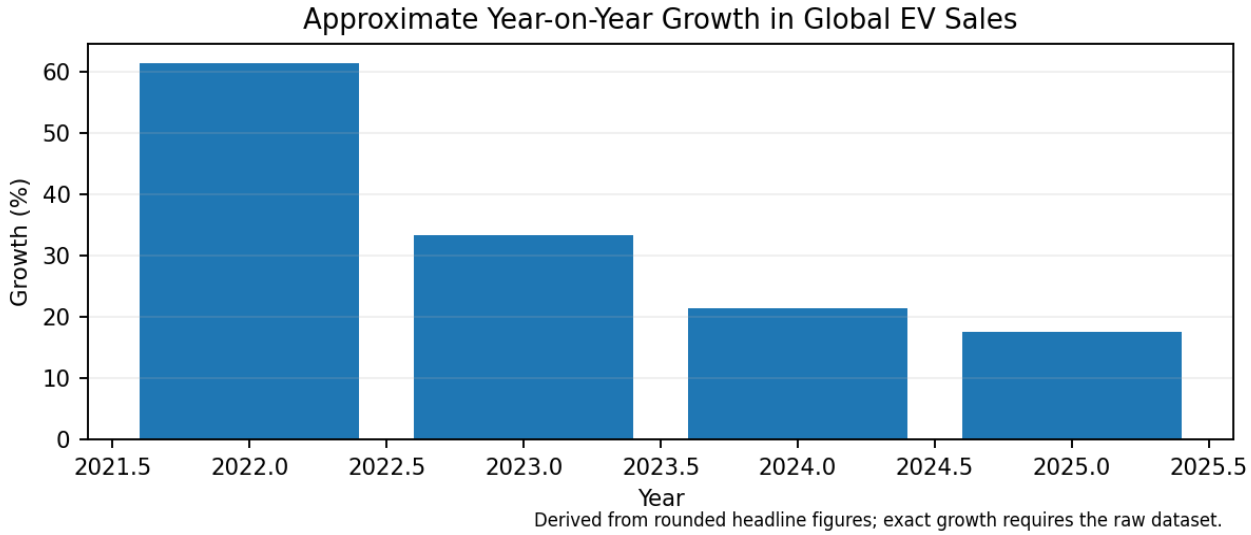


Figure 4. Illustrative year-on-year growth calculated from rounded global sales headline figures. Recalculate growth using the precise annual values from the selected source dataset.

DATA EXPORT AND REPRODUCIBILITY

After cleaning, the analysis dataset will be exported for dashboard development. A suitable filename is:

global_ev_analytics_clean.csv

The final dataset documentation should record the source URL, download date, selected indicators, units, filters, any removed duplicate rows, missing-value treatment, and the final number of records and columns. Keep an untouched copy of the original file so that all transformations can be reproduced.

PROPOSED OUTPUT FILES

File	Purpose
global_ev_analytics_clean.csv	Cleaned data for analysis and dashboards
ev_analysis.ipynb	Python cleaning, EDA, calculations, and charts
ev_sales_trend.png	Annual sales trend visualization
ev_country_comparison.png	Country or region ranking chart
Power BI dashboard	Interactive KPI cards, trends, and slicers
Tableau dashboard	Geographic and temporal EV exploration

The dashboard screenshots and final figures should be captured after the source dataset has been imported and the visuals have been validated. This keeps the report consistent with the actual implementation.

POWER BI OUTPUT

The cleaned dataset will be imported into Power BI to create an interactive Global EV Intelligence Dashboard. The dashboard is intended to help users compare adoption trends, sales, stock, and charging infrastructure by year and geography.

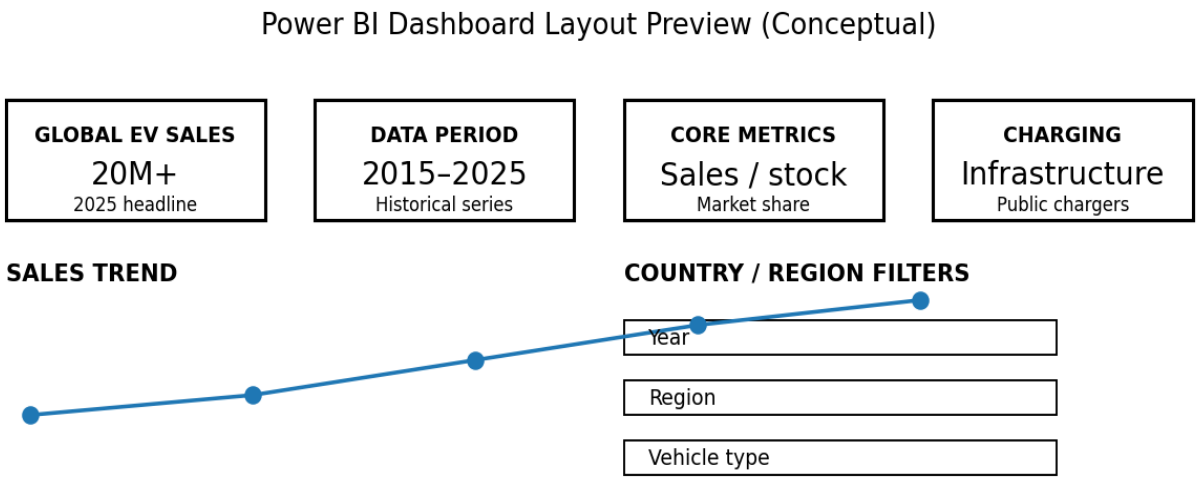


Figure 5. Conceptual dashboard layout preview. Replace this layout preview with a screenshot of the completed Power BI dashboard after implementation.

PROPOSED KPI CARDS AND VISUALS

- Total electric-car sales for the selected year and geography
- EV stock and EV sales share, where the source supports these measures
- Yearly sales trend and year-on-year growth
- Country or region ranking by sales or stock
- Charging infrastructure trend
- Slicers for year, region, country, and indicator

Power BI measures must use compatible indicators and units. Avoid summing overlapping regional aggregates together with their component countries.

TABLEAU OUTPUT

The cleaned data will also be used in Tableau to create an interactive EV Adoption Explorer. A map can be included if the dataset provides consistent geographic identifiers; otherwise, a ranked bar chart and regional heatmap will be used.

PROPOSED TABLEAU VIEWS

- Geographic map of EV sales, stock, or market share
- Year-wise EV adoption trend
- Country or region comparison chart
- Country-year heatmap
- Filters for year, geography, and indicator

RESULT

The project is designed to transform published electric-vehicle data into a cleaned analytical dataset and interactive dashboards using Python, Power BI, and Tableau. The analysis will provide a structured view of EV adoption trends, regional differences, and charging infrastructure. Final numerical conclusions will be stated only after the downloaded dataset has been cleaned, analyzed, and validated.

DATA SOURCE AND REFERENCES

1. International Energy Agency (IEA), Global EV Data Explorer: https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer

2. International Energy Agency (IEA), Global EV Outlook 2026 data product: https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2026

3. International Energy Agency (IEA), Global EV Outlook 2025, charging infrastructure analysis: https://www.iea.org/reports/global-ev-outlook-2025

Attribution: IEA data should be cited according to the licence and terms displayed on the exact dataset download page. Confirm permissions before redistribution.